Alejandro Medina Fernandez

Software Engineer
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Portfolio | **Linkedin**

Education

Universit degli Studi — Salerno, Italia
Ingegneria Informatica Bachelor's — 2024 - 2025

Polytechnic School of Engineering — Asturias, Spain
Information Technology Engineering Bachelor's — 2021 - 2026

Alfonso X el Sabio University — Madrid, Spain
Artificial Intelligence Mster's — 2026 - 2027

Experience

Indra Sistemas | **Software Engineer** — Asturias, Spain | Enero 2026 - Julio 2026
Development of tools for ATM (Air Traffic Management) systems.

Skills

Programming Languages:	C++, Java, Python, C
Libraries/Frameworks:	React, TypeScript, Tailwind CSS
Tools / Platforms:	Git, Jenkins, SonarQube, Docker, JaCoCo, FindSecBugs, PiTest, Bitbucket, MCPs, LLMs
Databases:	MongoDB, MariaDB, PostgreSQL

Projects / Open-Source

commons-dbutils | **Link** — *HTML, Java, JavaScript, CSS, Shell*
Contribution to the Apache Commons DbUtils project focused on improving quality with tests generated using Randoop, coverage analysis, and JMH benchmarking. The work includes security reports, documentation, and adjustments to the Maven build process.

cleaningrobotpy | **Link** — *Python*
Autonomous cleaning robot compatible with Raspberry Pi that models motors, infrared sensors, and a battery system to react to obstacles and dirty areas. The Route Management System coordinates the route on a grid and allows for scenario simulation with automated tests.

QR-generator | **Link** — *CSS, Python, JavaScript, HTML*
Responsive web application that generates QR codes both from URLs and from uploaded images hosted on Imgur. It is built solely with HTML, CSS, and JavaScript, and integrates lightweight animations and status messages for a clear experience.

Password-manager | **Link** — *Python*
Python password manager with a Tkinter interface that encrypts credentials into JWT tokens and persists them in PostgreSQL. Includes the loading of external SQL queries, input masking, and utilities to import or reveal stored tokens.

RemoteWordle | **Link** — *Java*
Competitive Wordle game with a Java client-server architecture that synchronizes matches in real time, manages multiple connections, and records each attempt in logs for analysis. It includes a custom dictionary, alias control, and coordinated restarts from the server.